Verification and Validation Report: Software Engineering

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1 Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

2 Symbols, Abbreviations and Acronyms

symbol	description
T	Test

[symbols, abbreviations or acronyms – you can reference the SRS tables if needed —SS]

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3 System evaluation

3.1 New user account creation

Table 1: Software Test Cases for User Account Creation

Test Case ID	**user_acc_create**
Control	Automatic
Rationale	The system must be able to create and store a new account
State	User does not have an account on the application.
Input	Govt id (name, date of birth), license number (PT) OR invite code (patient), email, password (new)
Output	'Account created successfully' displayed on page and user is successfully added to the user database
How the test will be performed	The test will be conducted with the use of a script which will input valid, invalid and edge case values
Test Case ID	**Invalid_formatCreate**
Control	Automatic
Rationale	The data input by the user has to be in the correct format for our system to accept it as valid.
State	User does not have an account on the application.
Input	Incorrect format of license number/invite code OR incorrect format of date of birth, email, password(new)
Output	System displays 'incorrect format of one or more inputs'
How the test will be performed	The test will be conducted with the use of a script which will input valid, invalid and edge case values
Test Case ID	**Invalid_sign_up_creds**
Control	Automatic

Rationale	Successful verification for both types of users is needed to use the application
State	User does not have an account on the application.
Input	Incorrect license number/mismatch of license number and govt id OR incorrect format of invite code, email, password(new)
Output	System displays ’License number cannot verified’ OR ’invalid invite code’
How the test will be performed	The test will be conducted with the use of a script which will input valid, invalid and edge case values
Test Case ID	**Acc_exists**
Control	Automatic
Rationale	New users should not already have an account on the system
State	Account already exists on system
Input	Govt id (name, date of birth), license number/invite code, email, password
Output	’Account already exists. Wrong password. Try again.’
How the test will be performed	The test will be conducted with the use of a script which will input valid, invalid and edge case values

3.2 PT focused testing

Table 2: Software Test Cases for Physical Therapist (PT) Functions

Test Case ID	**PT_deletes_patient_acc**
Control	Automatic
Rationale	This ensures PTs control over patients’ accounts. PT can remove patient(s) profiles from the application.
State	PT and patient have account on application, PT has the patient on their patient list

Input	PT requests a specific patient's account deletion
Output	Account deleted from user database and PT patient list. Patient is unable to log in ('Account not found').
How the test will be performed	The test will be conducted with the use of a script which will input valid, invalid and edge case values
Test Case ID	**PT_adjusts_patient_exercise**
Control	Automatic
Rationale	The system allows the PT to obtain an overview of patient performance and adjust different parameters accordingly (eg. number of repetitions).
State	PT and patient have an account on application, PT has patient on their patient list
Input	PT accesses the patient's profile page and adds/modifies exercise
Output	Patient exercise criteria is modified.
How the test will be performed	The test will be conducted with the use of a script which will input valid, invalid and edge case values
Test Case ID	**PT_sends_invite**
Control	Automatic
Rationale	A patient can only sign up if they have an invite code from the PT.
State	Patient does not have an account on the application while PT does.
Input	PT adds patient's information and contact method (where the code will be sent). Contact methods may include: email, text messages, and other forms of communication.
Output	An 'Invite sent' message is displayed on the screen
How the test will be performed	The test will be conducted with the use of a script which will input valid, invalid and edge case values

3.3 System specific testing

Table 3: Software Test Cases for Exercise System and Additional Functions

Test Case ID	**System_exercise_instruct**
Control	Manual
Rationale	In order to record the patient performing the exercise, the system has to be able to instruct the patient on how to perform it.
State	Patient is logged into the application and has it open on their device. Patient has selected the exercise they want to perform on their exercise list. Once selected, system displays an overview of the steps required.
Input	Patient clicks on 'start'
Output	System displays (text or audio) instructions on how to perform the selected exercise
How the test will be performed	• Patient navigates to 'Exercise list' • Patient clicks on the exercise they would like to perform • The system displays the instructions (audio/text) OR system guides the exercise performance by the patient
Test Case ID	**System_record_exercise**
Control	Manual
Rationale	Patient performance of the exercise has to be successfully recorded to aid system analysis on its correctness.
State	Patient is currently logged in and has the application open on their device. Patient has selected the exercise they would like to perform from the exercise list.
Input	Patient clicks on the 'record' button
Output	Exercise successfully recorded, saved and ready for viewing

How the test will be performed	• Patient clicks on the 'exercise list' and selects the exercise they would like to perform • System directs patient to a screen with a live video stream of themselves • Patient clicks on the 'record' button
Test Case ID	**System_analysis_feedback**
Control	Manual
Rationale	System has to effectively perform analysis on patient exercise form to provide feedback.
State	Patient is recording exercise
Input	System receives visual of patient performing exercise
Output	System detects intended deviations and correctly alerts (audio and visual) the user
How the test will be performed	• Patient performs exercise with intentional form deviations • System monitors exercise performance in real-time • Verify system generates appropriate audio and visual alerts for detected deviations
Test Case ID	**Video_store_retrieval**
Control	Manual
Rationale	The system has to be able to retrieve the most recent recorded video in storage (either locally or database).
State	Patient has performed and recorded exercise being performed
Input	Patient indicates where video should be stored (locally or database).
Output	Video has been successfully stored and a pop-up ('Saved successfully') is displayed

How the test will be performed	- Patient completes exercise recording - Patient selects storage location (local or database) - System saves the video and displays confirmation message - Verify video can be retrieved from the selected storage location
Test Case ID	**post_ex_patient_feedback**
Control	Automatic
Rationale	The system has to be able to store patient feedback of the performed exercise (e.g. the level of (perceived) difficulty of the exercise). See FR 9.
State	Patient has performed and recorded exercise being performed and system has successfully performed analysis of the performance
Input	Patient feedback (See Appendix).
Output	Answers are stored successfully
How the test will be performed	- System displays feedback form after exercise completion - Patient provides feedback on exercise difficulty and other metrics - System stores the feedback data - Verify feedback is accessible for review by PT
Test Case ID	**Incorrect_form_adjustment**
Control	Manual
Rationale	The system has to be able to differentiate between correct and incorrect forms to provide accurate feedback.
State	Patient is performing the selected exercise.
Input	Incorrect form during exercise performance
Output	Adjustment directives displayed to patient

How the test will be performed	• Patient performs exercise with incorrect form • System detects the incorrect posture/movement • System provides real-time adjustment instructions • Verify patient can understand and implement the corrections
Test Case ID	**System_rejects_duplicate**
Control	Automatic
Rationale	The system will traverse all login credentials stored on the database and will disallow account creation if a duplicate identifying credential exists (e.g. duplicate email, or user login ID). Note: This is a design detail that will be decided later in the project timeline.
State	User has input information in all of the required fields on the 'Create Account' page.
Input	User clicks the 'Submit' button
Output	A pop-up with the message 'Account already exists' (Note: The team has not decided what page to direct to after this message appears)
How the test will be performed	• User attempts to create account with existing email/login ID • System checks database for duplicates • System displays error message and prevents account creation • Verify existing accounts remain unaffected

4 Nonfunctional Requirements Evaluation

Table 4: Non-Functional Requirements Testing

Test Case ID	**ui_testing**

Type	Static
Rationale	Ensure user interface meets usability standards and stakeholder expectations
State	N/A
Input	User interface elements and survey responses
Output	Modified user interface based on feedback
How the test will be performed	• A survey will be given to users (see Stakeholders) to test different user interface elements • See Appendix for survey questions • Based on the feedback received, the user interface will be modified in order to satisfy additional requirements
Test Case ID	**ex_int_speed**
Type	Dynamic
Rationale	Ensure exercise interface loads within acceptable time limits for good user experience
State	User is logged in and has application open on their device
Input	User clicks on exercise interface
Output	Interface loads within 2 seconds
How the test will be performed	• Multiple users will access the exercise interface under normal network conditions • Load times will be measured and recorded • Statistical analysis will determine if 95% of loads occur within 2 seconds
Test Case ID	**fail_recovery**
Type	Dynamic
Rationale	The system must be able to restore its pre-failure state 95% of the time within 5 minutes
State	User is currently using the application on their device

Input	System failure (simulated)
Output	System restarts and recovers within 5 minutes
How the test will be performed	• Test will be repeated numerous times with various failure scenarios • Recovery time and success rate will be measured • System will be forced to fail during different operations • Verify data integrity after recovery
Test Case ID	**Interrupted_vid**
Type	Dynamic
Rationale	The system must be able to attempt resubmission of video after restoring internet connection
State	A video of the user performing the exercise is being uploaded
Input	Internet connection interruption during upload
Output	Video is successfully re-queued for upload to the system. Re-upload is resumed at the point of failure
How the test will be performed	• Simulate network disconnection during video upload • Restore network connection after interruption • Verify system detects interruption and queues video for re-upload • Confirm upload resumes from point of failure (not restarting) • Validate video integrity after successful upload
Test Case ID	**PR-RFT 3, PR-RFT 4, PR-CR1, PR-SER 1**
Type	N/A
Rationale	These requirements address advanced system capabilities beyond current testing scope
State	N/A
Input	N/A

Output	N/A
How the test will be performed	• These requirements are unable to be tested in a timely manner due to project constraints (time, money) • Testing deferred to future development phases • Requirements marked for re-evaluation in next project iteration

4.1 Usability

4.2 Performance

4.3 etc.

5 Comparison to Existing Implementation

This section will not be appropriate for every project.

6 Unit Testing

7 Changes Due to Testing

[This section should highlight how feedback from the users and from the supervisor (when one exists) shaped the final product. In particular the feedback from the Rev 0 demo to the supervisor (or to potential users) should be highlighted. —SS]

8 Automated Testing

9 Trace to Requirements

10 Trace to Modules

11 Code Coverage Metrics

Appendix — Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Reflection.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which parts of this document stemmed from speaking to your client(s) or a proxy (e.g. your peers)? Which ones were not, and why?
4. In what ways was the Verification and Validation (VnV) Plan different from the activities that were actually conducted for VnV? If there were differences, what changes required the modification in the plan? Why did these changes occur? Would you be able to anticipate these changes in future projects? If there weren't any differences, how was your team able to clearly predict a feasible amount of effort and the right tasks needed to build the evidence that demonstrates the required quality? (It is expected that most teams will have had to deviate from their original VnV Plan.)